

NGUYEN LUONG, PhD

Research Software Engineer | Machine Learning & Scientific Computing

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Research Software Engineer and PhD in Computer Science working across scientific computing, machine learning, digital health, and production software. Develops reproducible ML/HPC workflows, containerized research environments, and training/evaluation pipelines at ELLIS Institute. Research expertise in longitudinal behavioral sensing, statistical modeling, time-series analysis, and multimodal sensor data.

TECHNICAL SKILLS

ML / Data Science	Machine Learning, Deep Learning, Statistical Modeling
HPC / Workflows	HPC/Slurm, GPU computing, Snakemake, Apptainer
Frameworks	PyTorch, PyTorch Lightning, scikit-learn, XGBoost, LightGBM
Profiling	PyTorch Profiler, NVIDIA Nsight Systems, py-spy, cProfile
Languages	Python, R, SQL, Swift
Data Engineering	Pandas, Polars, DuckDB, Apache Spark (PySpark), Parquet
MLOps / Infra	Docker, Kubernetes, Ansible, CI/CD, MLflow, Weights & Biases, Git

EXPERIENCE

Research Software Engineer | Aalto University & ELLIS Institute, Finland Nov 2025 - Present

Aalto Scientific Computing, Department of Computer Science

- ▶ Design and support reproducible ML/HPC workflows across research groups, from experimental design and model selection to training and evaluation pipeline architecture
- ▶ Build containerized GPU/HPC environments with Docker and Apptainer, and optimize workloads for Slurm clusters and shared accelerator infrastructure
- ▶ Maintain research software engineering workflows with Git, CI/CD, automated testing, documentation, packaging, and PyPI releases
- ▶ Occasional instructor with CodeRefinery, a Nordic training network teaching researchers essential tools for reproducible, reusable, and open software and data workflows.

Doctoral Researcher | Aalto University, Finland

Nov 2021 - Sep 2026

Advisor: Dr. Talayeh Aledavood

- ▶ Developed statistical and machine learning methods to characterize daily routines and mental-health-related behavior from longitudinal smartphone and wearable data across multiple cohorts
- ▶ Integrated multimodal behavioral signals including sleep, physical activity, mobility, communication, and device use, with an emphasis on reproducible and cross-cohort analysis
- ▶ 9 peer-reviewed papers across digital health, medical informatics, and research software, including JAMIA, JMIR, EPJ Data Science, and SoftwareX
- ▶ Contributed to Niimpy, an open-source Python toolkit for reproducible preprocessing and feature extraction from smartphone sensing data (SoftwareX, 2023)

Leading parking application in Finland

- Owned end-to-end iOS engineering for a consumer parking platform with 200k+ monthly active users, driving technical architecture, product delivery, release planning, and App Store operation
- Improved user retention and reduced churn by 10% through analytics-driven product iteration, funnel analysis, and controlled A/B testing of key user journeys.
- Improved release cycle from monthly to bi-weekly by designing a CI/CD and automated testing workflow, reducing manual release overhead and enabling faster and safer delivery.
- Led a large-scale modernization of the iOS codebase from Objective-C to Swift, restructuring legacy components to improve maintainability and application stability.

EDUCATION

PhD in Computer Science | Aalto University, Finland

2026

Dissertation: Behavioral Sensing for Routine Characterization and Mental Health

- Advisor: Dr. Talayeh Aledavood

MSc in Data Science, Machine Learning, and AI | Aalto University, Finland

2021

Graduated with honors

- Thesis: Estimating treatment effects from clinical data using a Transformer-based model
- Advisor: Prof. Pekka Marttinen

BSc in Computer Science | Metropolia AMK, Finland

2017